FastTrack

Phase 2

Team 3

Member Name	Member Roll #	Primary Responsibility
Romesha Afzaal	24L-3001	Design Class Diagram
M. Hassan Ashraf	24L-3073	Design Class Diagram
Abdul Ahad Shams	24L-3029	Sequence Diagrams (UC-01 to UC-20)
Kabeer Ahmed Shahzeb	24L-3087	Sequence Diagrams (UC-21 to UC-39)
Syed Saad Ali	24L-2549	Package Diagram

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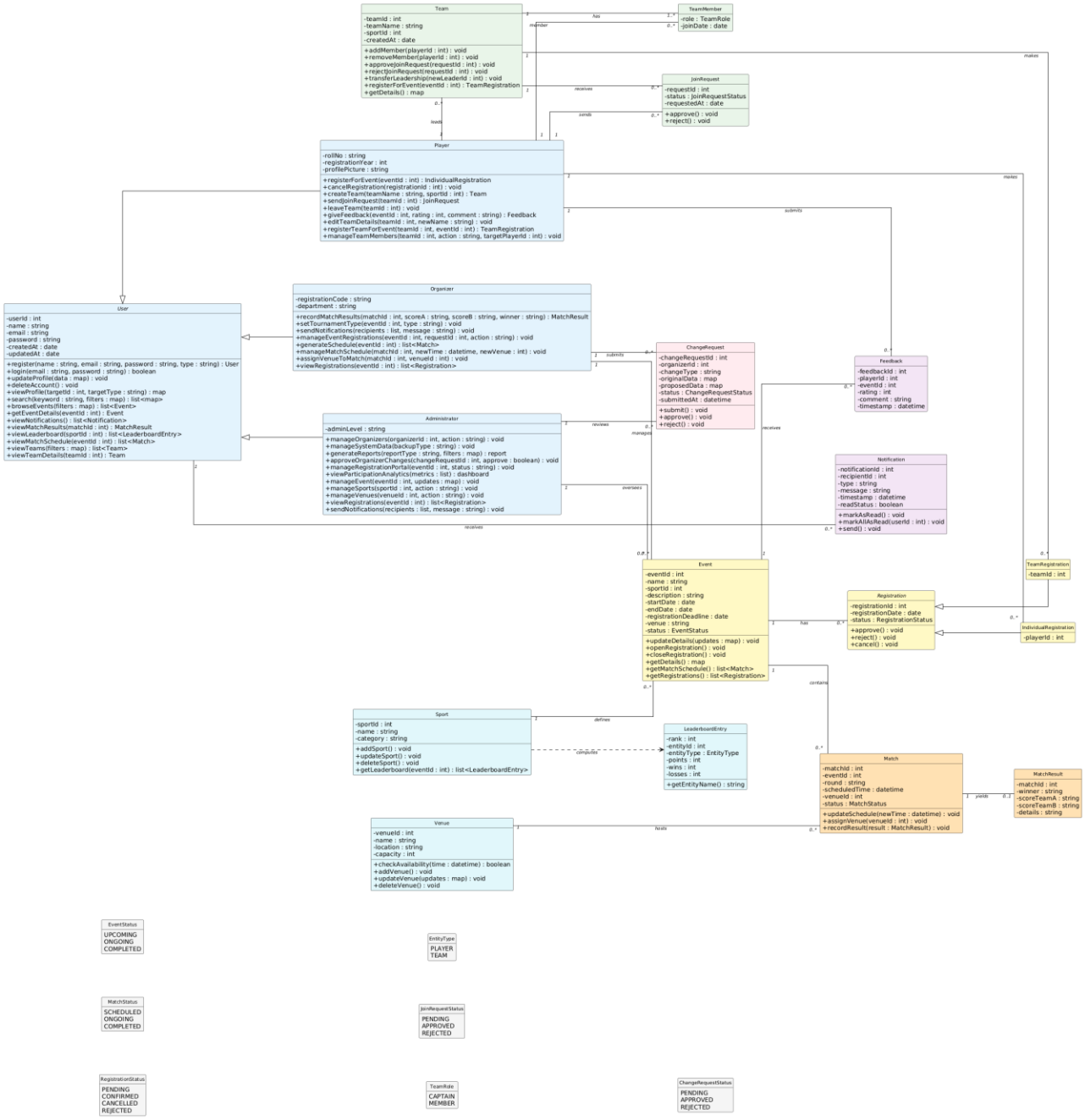
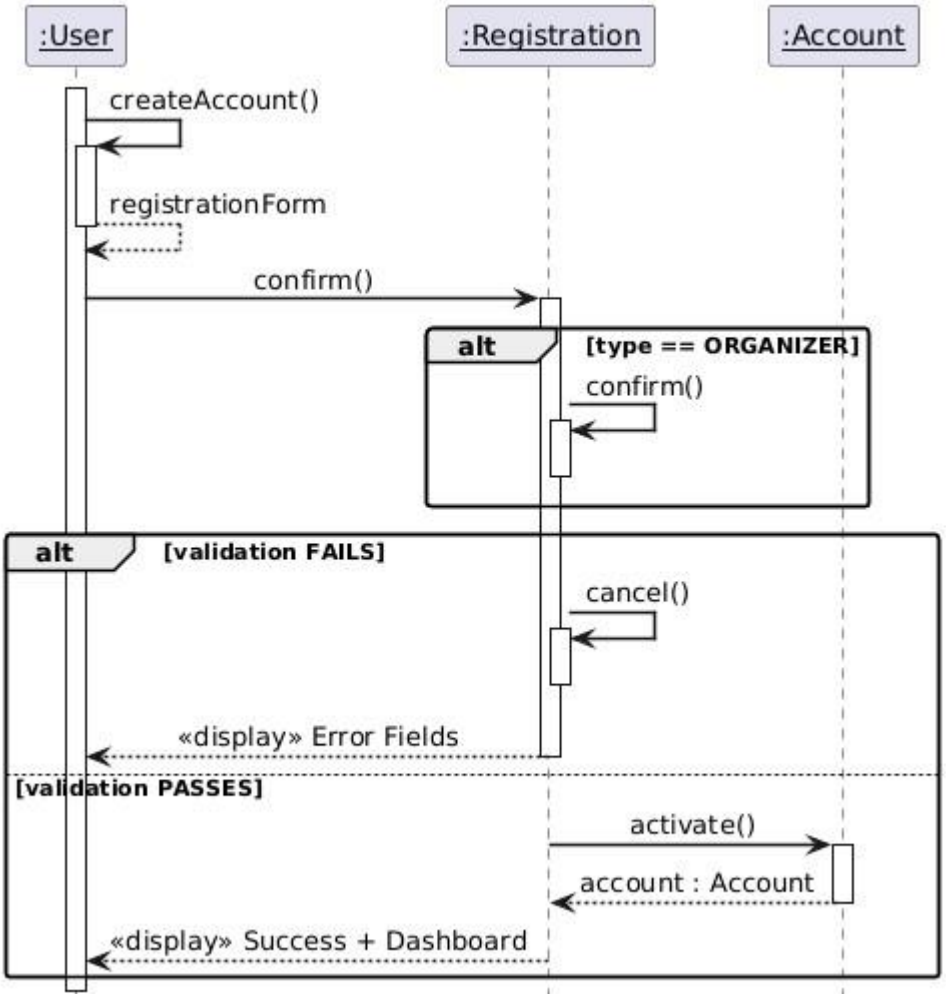
1. Design Class Diagram:

Note: Kindly zoom in to view clearly.

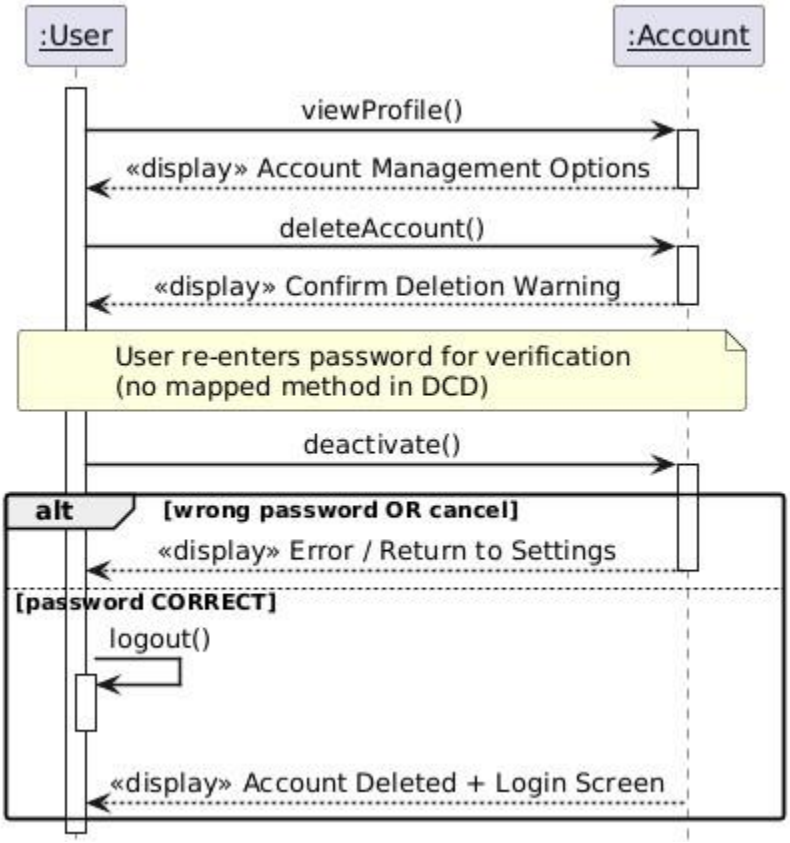


2. Sequence Diagrams:

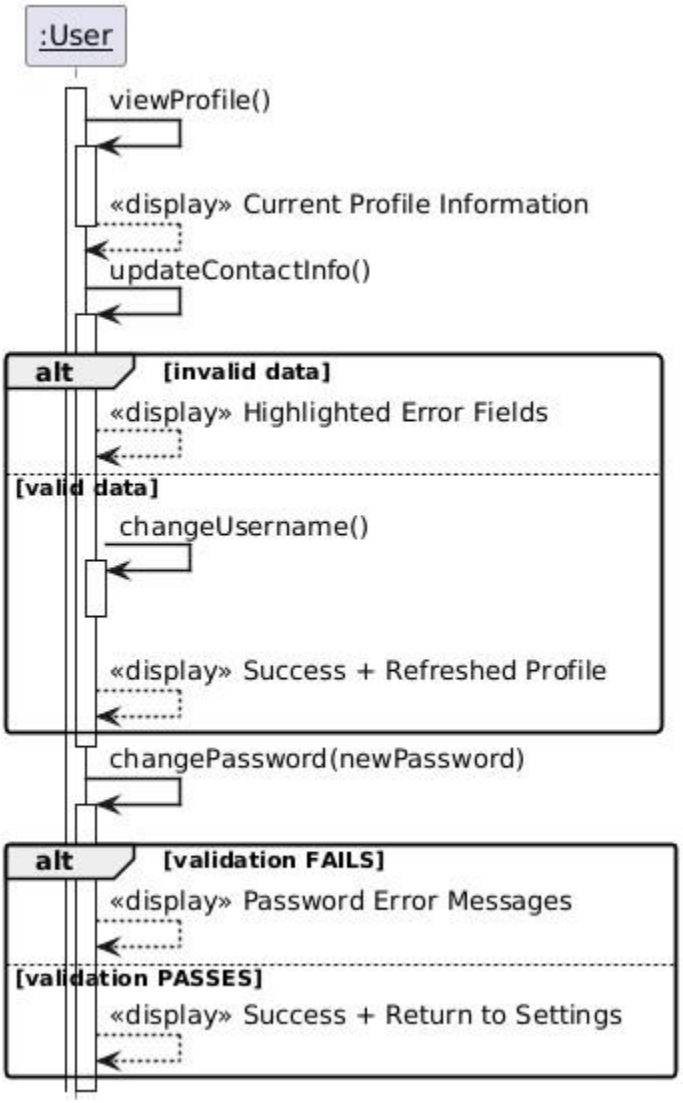
2.1 UC-01



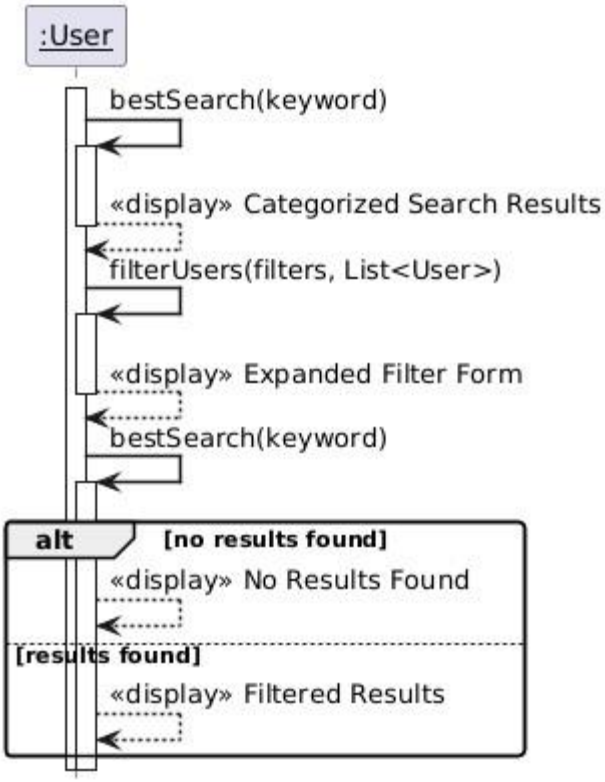
2.2 UC-02


:User
:Account
viewProfile()
«display» Account Management Options
deleteAccount()
«display» Confirm Deletion Warning
User re-enters password for verification
(no mapped method in DCD)
deactivate()
alt
[wrong password OR cancel]
«display» Error / Return to Settings
[password CORRECT]
logout()
«display» Account Deleted + Login Screen

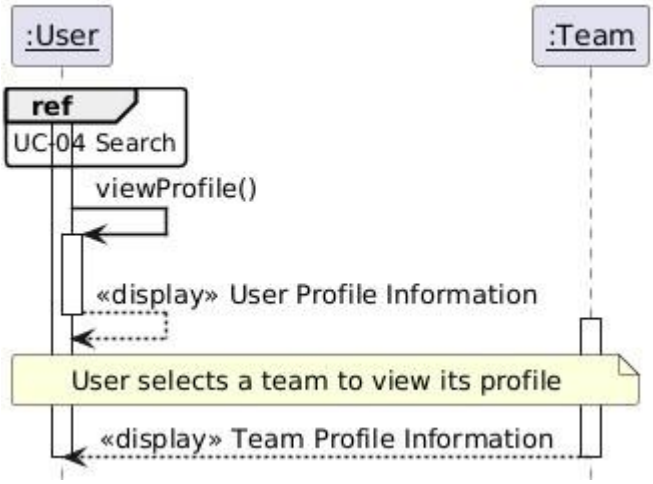
2.3 UC-03


:User
viewProfile()
«display» Current Profile Information
updateContactInfo()
alt
[invalid data]
«display» Highlighted Error Fields
[valid data]
changeUsername()
«display» Success + Refreshed Profile
changePassword(newPassword)
alt
[validation FAILS]
«display» Password Error Messages
[validation PASSES]
«display» Success + Return to Settings

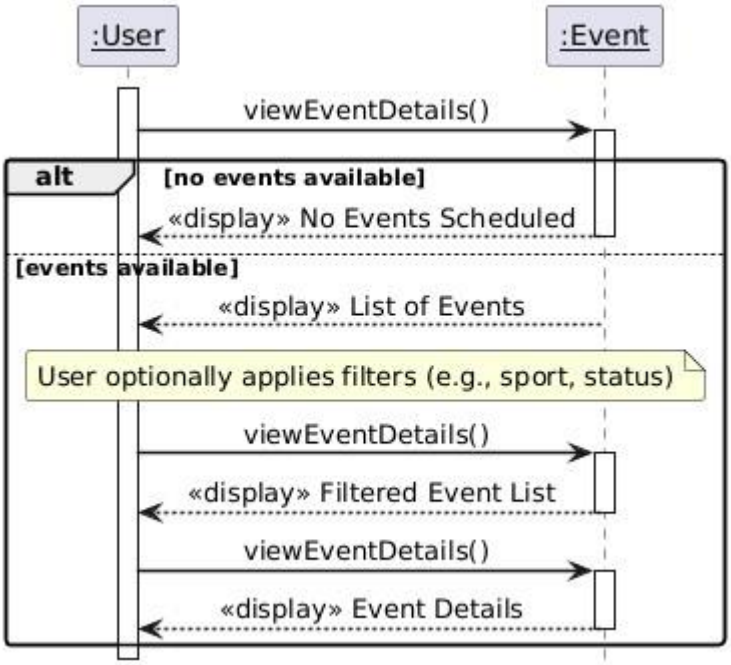
2.4 UC-04


:User
bestSearch(keyword)
«display» Categorized Search Results
filterUsers(filters, List<User>)
«display» Expanded Filter Form
bestSearch(keyword)
alt
[no results found]
«display» No Results Found
[results found]
«display» Filtered Results

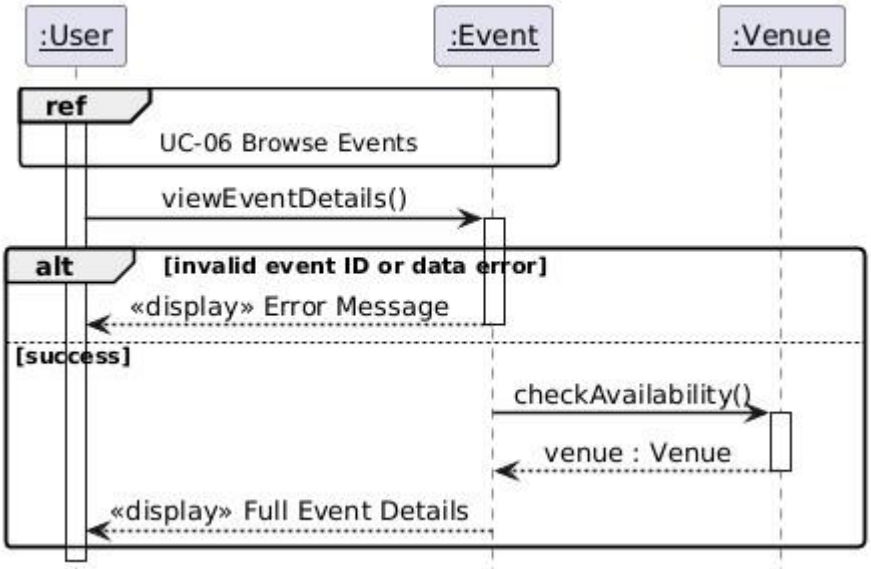
2.5 UC-05



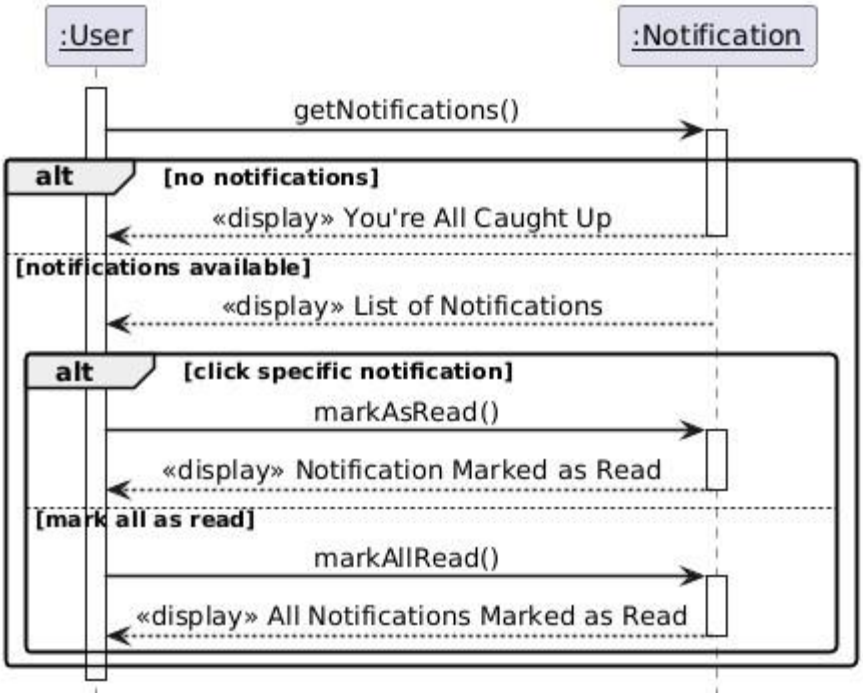
2.6 UC-06



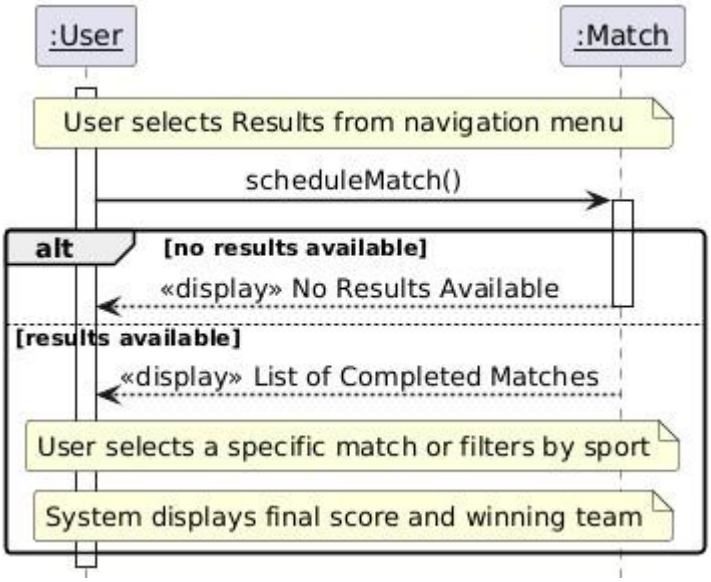
2.7 UC-07



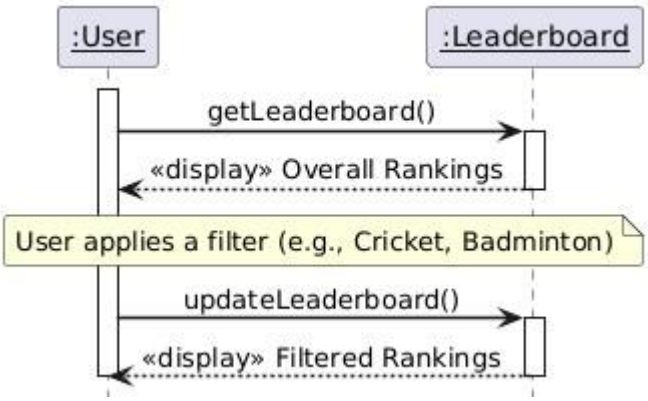
2.8 UC-08



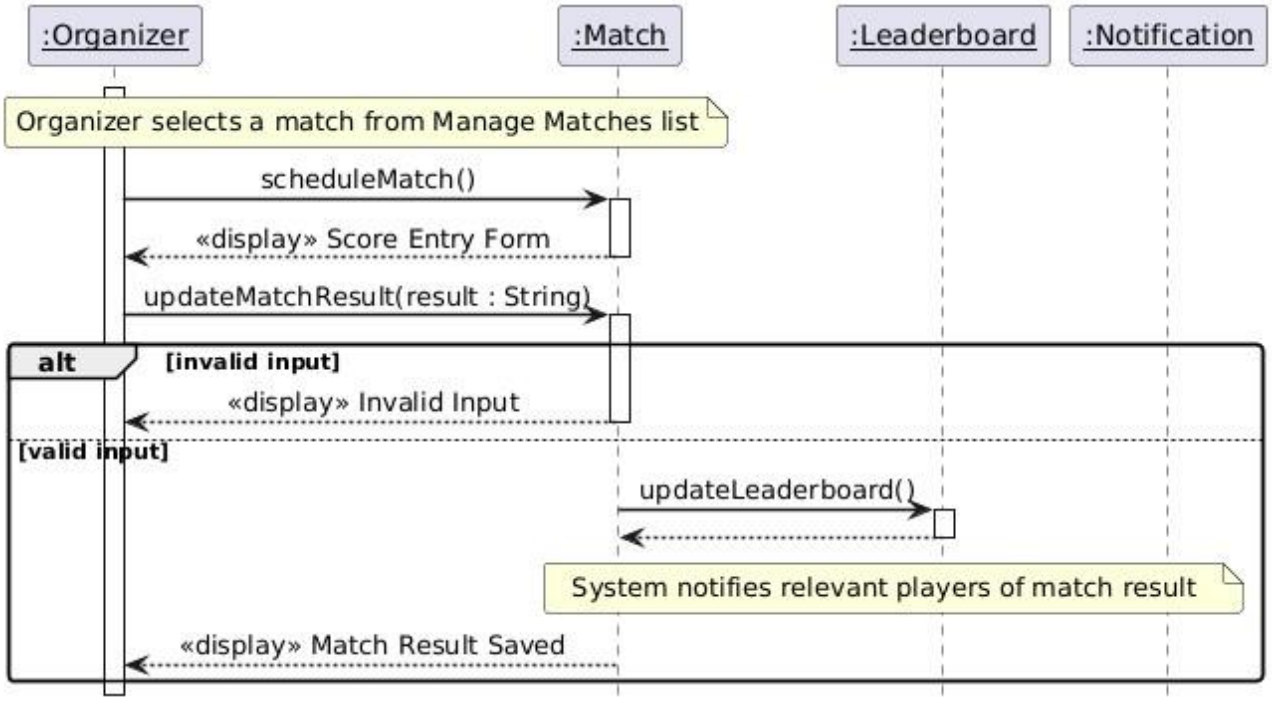
2.9 UC-09



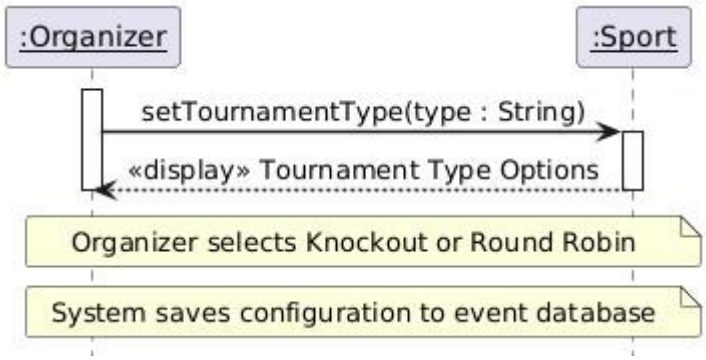
2.10 UC-10



2.11 UC-11

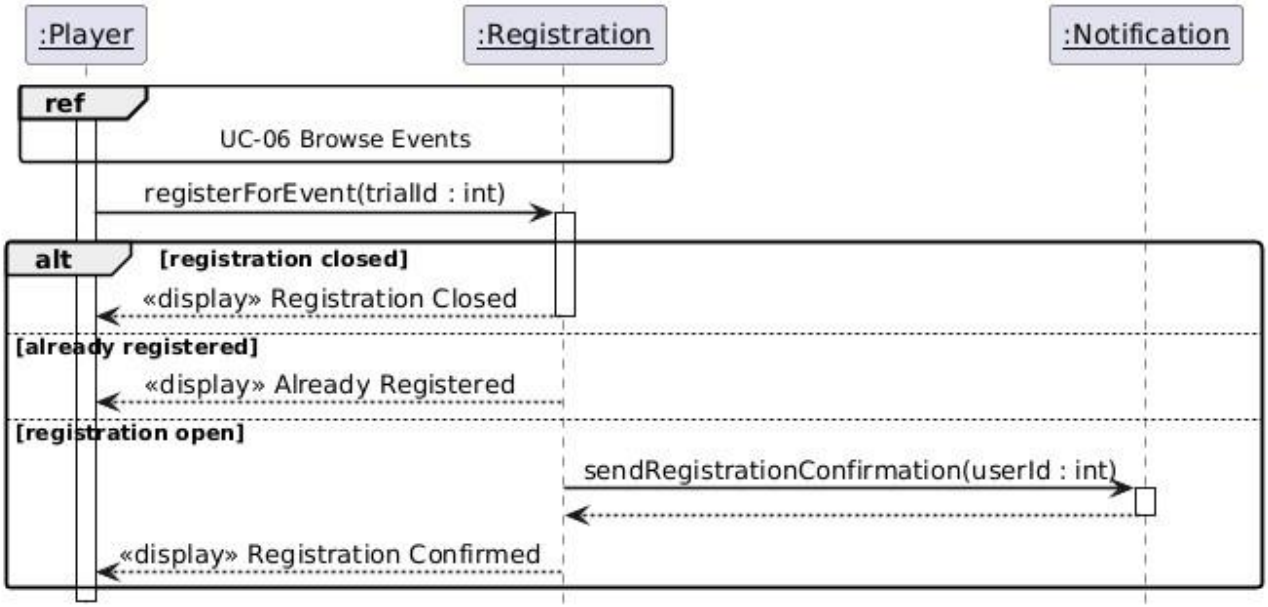


2.12 UC-12



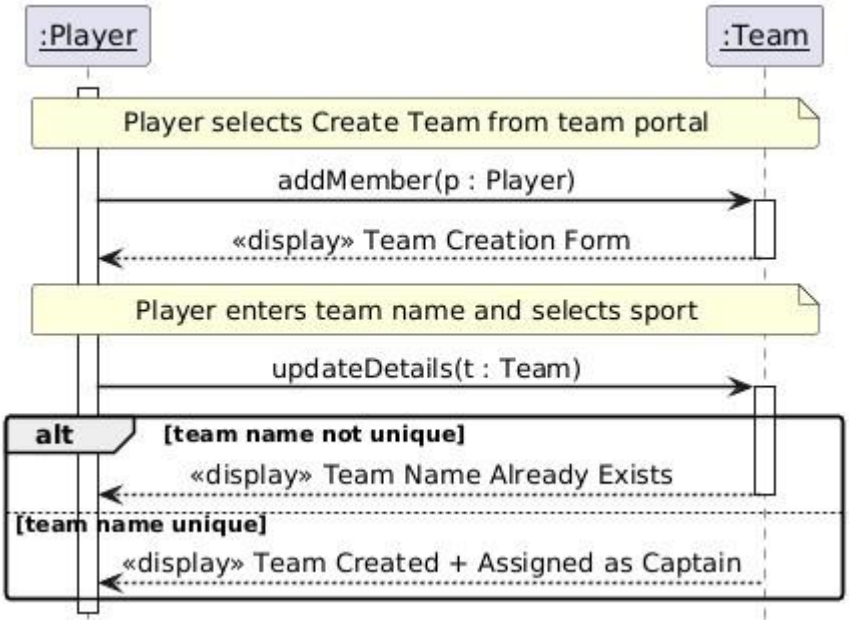
2.13 UC-13

2.14 UC-14

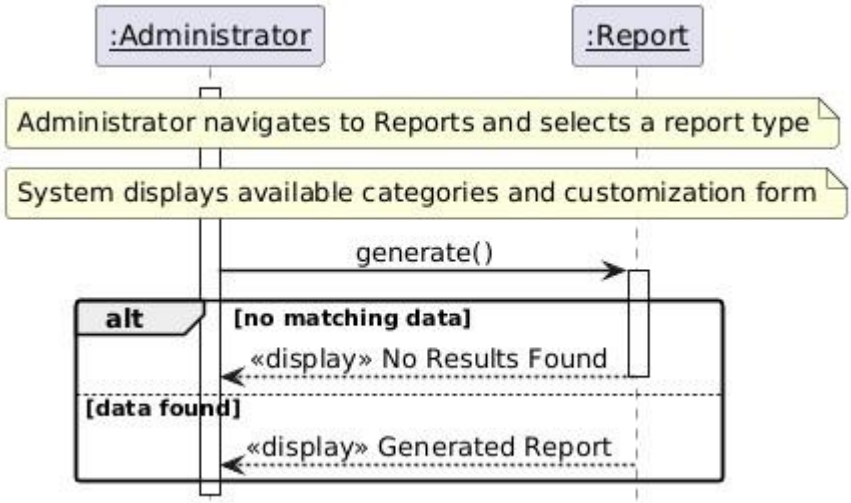


2.15 UC-15

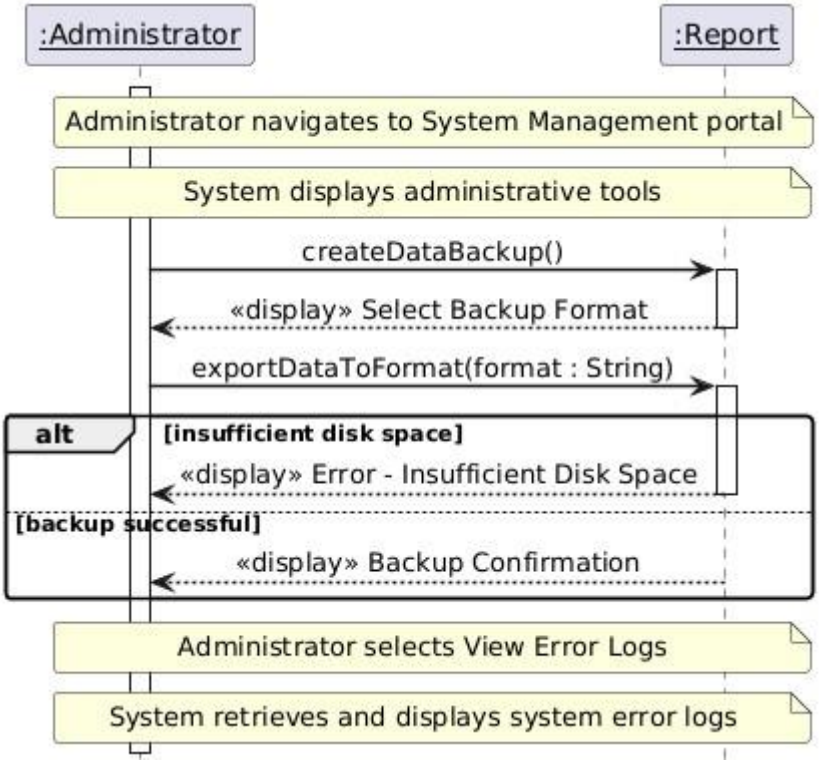
2.16 UC-16



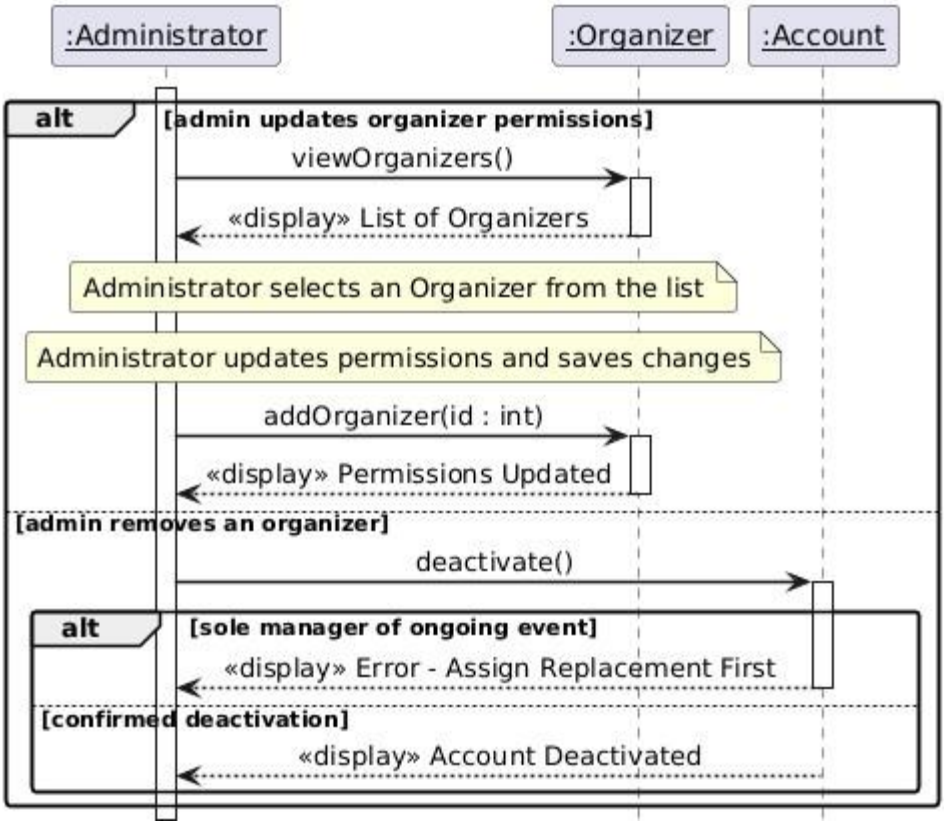
2.17 UC-17



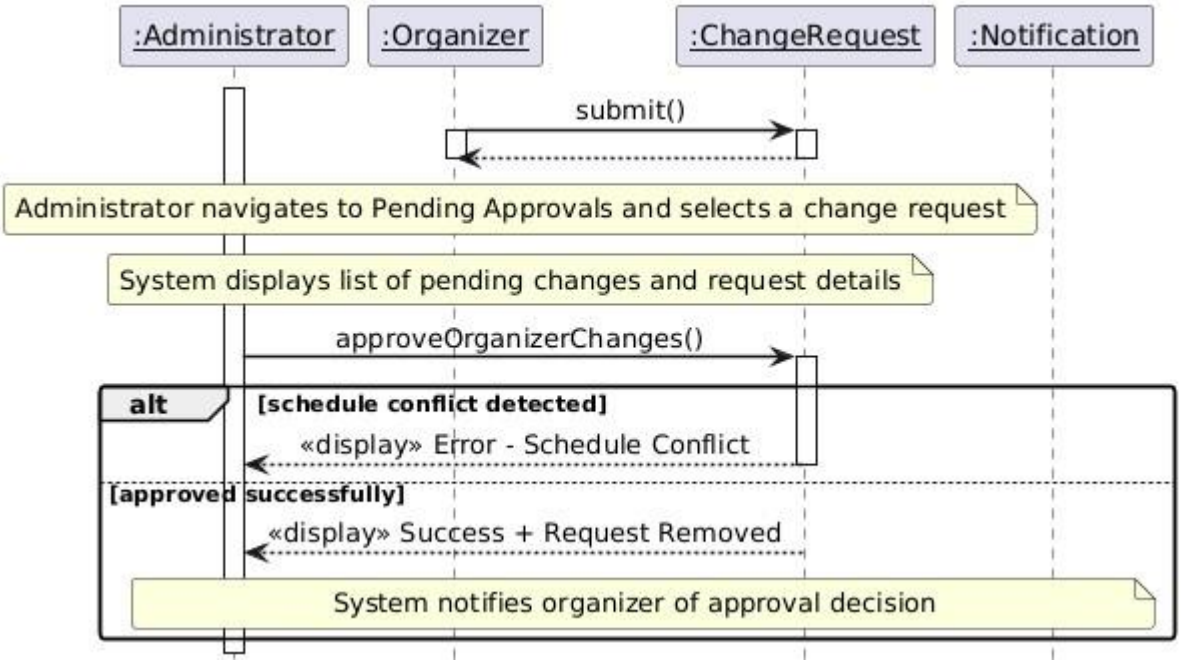
2.18 UC-18



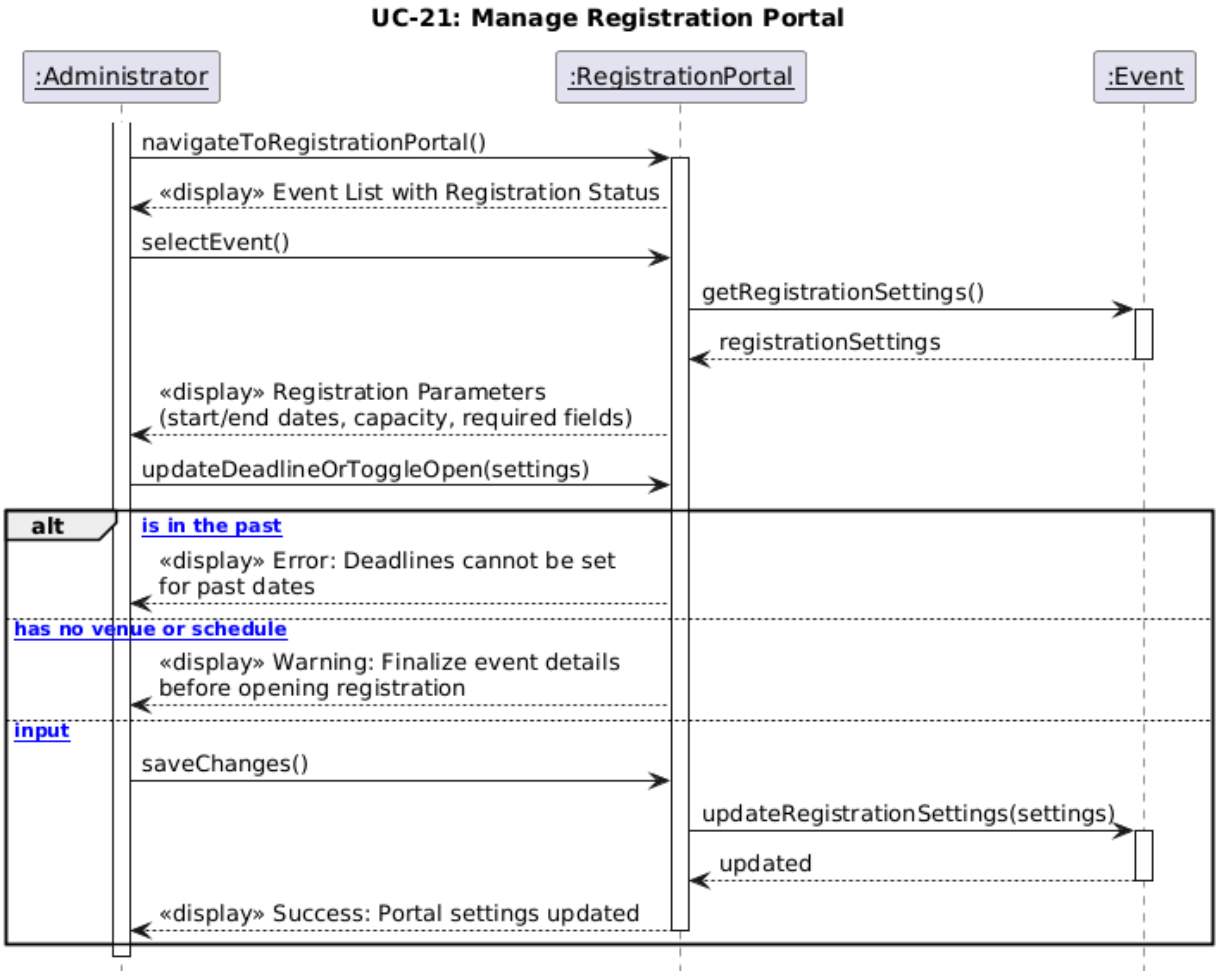
2.19 UC-19


:Administrator
:Organizer
:Account
alt
[admin updates organizer permissions]
viewOrganizers()
«display» List of Organizers
Administrator selects an Organizer from the list
Administrator updates permissions and saves changes
addOrganizer(id : int)
«display» Permissions Updated
[admin removes an organizer]
deactivate()
alt
[sole manager of ongoing event]
«display» Error - Assign Replacement First
[confirmed deactivation]
«display» Account Deactivated

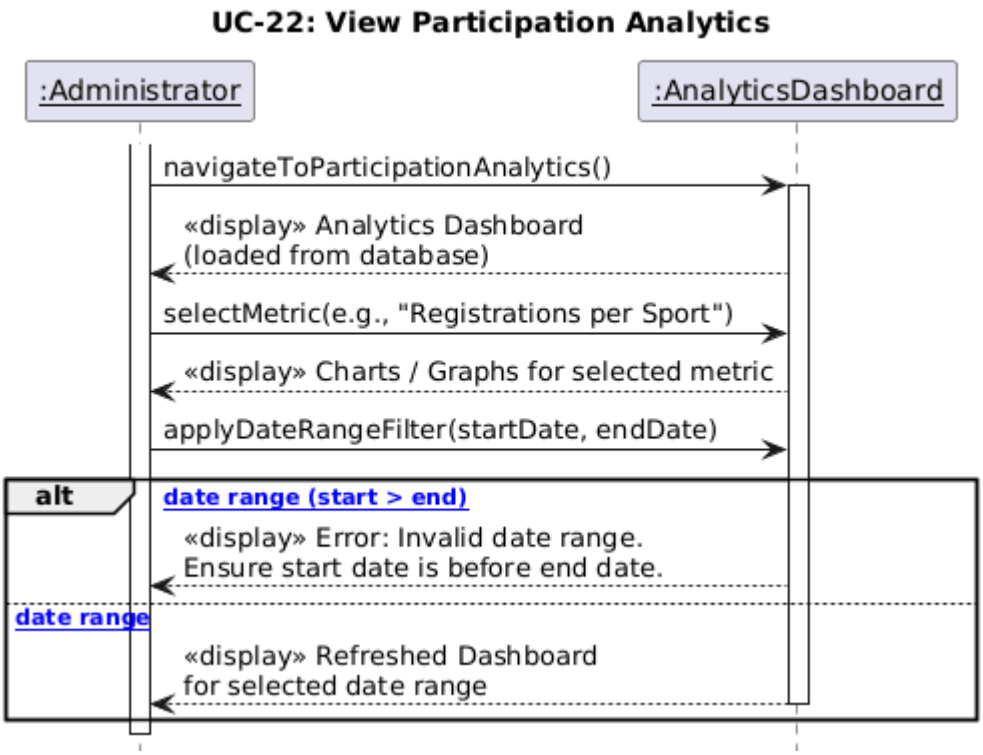
2.20 UC-20


:Administrator
:Organizer
:ChangeRequest
:Notification
submit()
Administrator navigates to Pending Approvals and selects a change request
System displays list of pending changes and request details
approveOrganizerChanges()
alt
[schedule conflict detected]
«display» Error - Schedule Conflict
[approved successfully]
«display» Success + Request Removed
System notifies organizer of approval decision

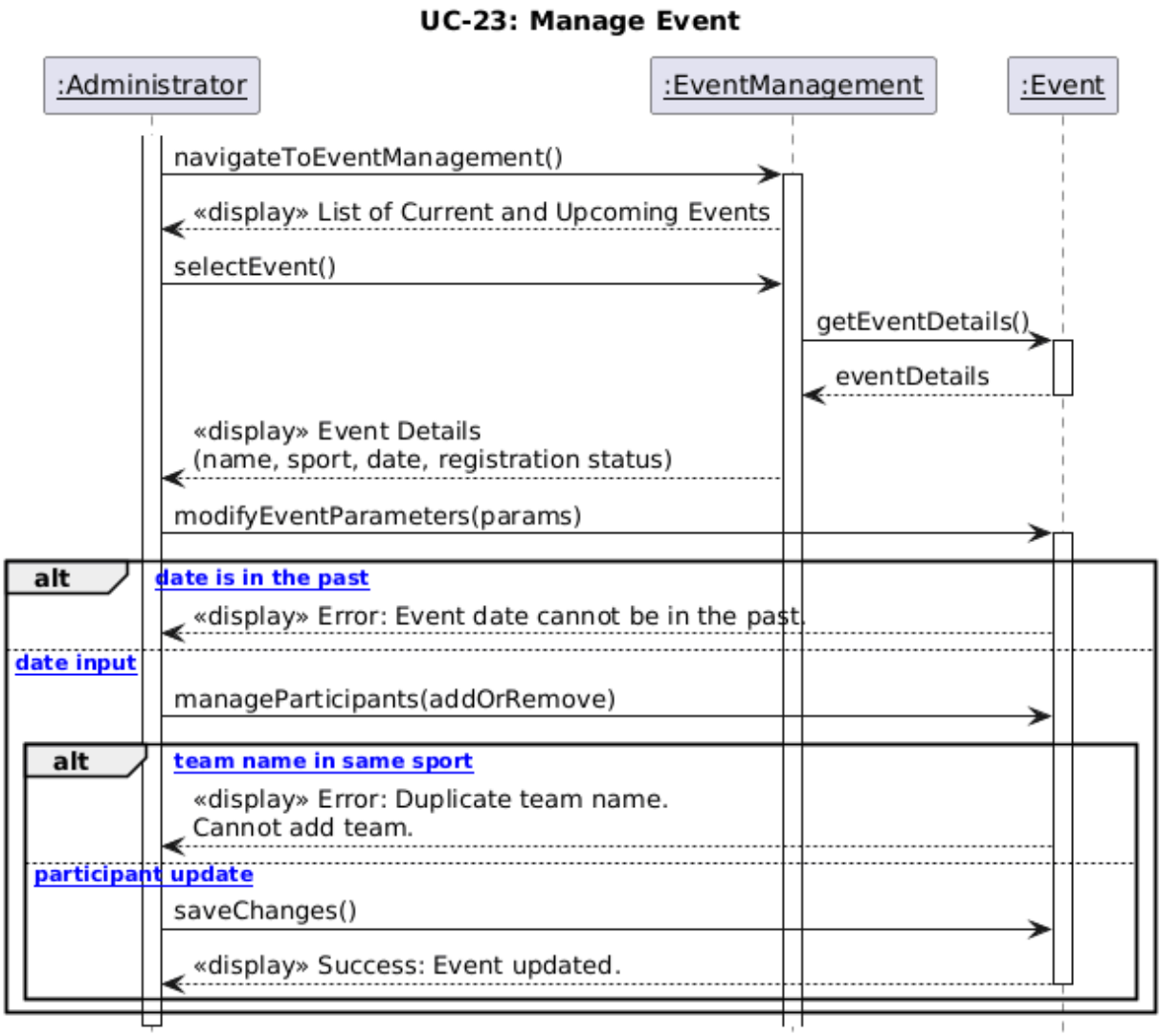
2.21 UC-21


UC-21: Manage Registration Portal
:Administrator
:RegistrationPortal
:Event
navigateToRegistrationPortal()
«display» Event List with Registration Status
selectEvent()
getRegistrationSettings()
registrationSettings
«display» Registration Parameters
(start/end dates, capacity, required fields)
updateDeadlineOrToggleOpen(settings)
alt
is in the past
«display» Error: Deadlines cannot be set
for past dates
has no venue or schedule
«display» Warning: Finalize event details
before opening registration
input
saveChanges()
updateRegistrationSettings(settings)
updated
«display» Success: Portal settings updated

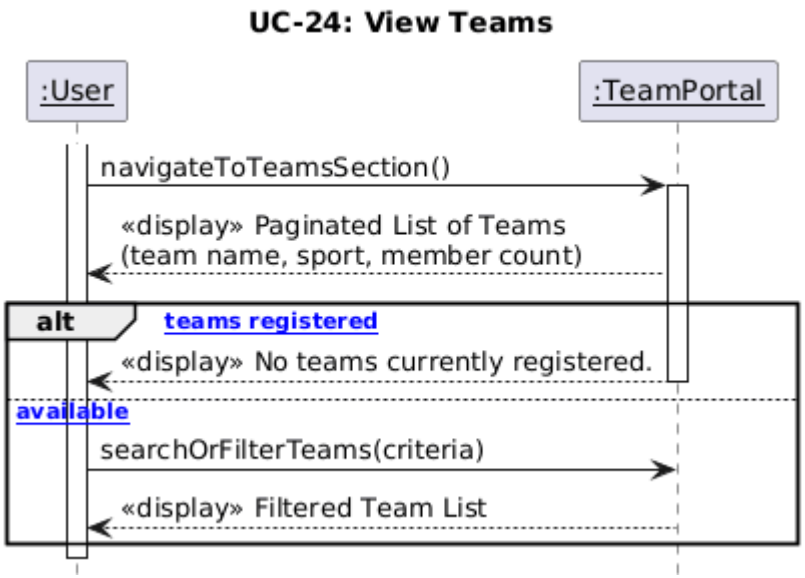
2.22 UC-22


UC-22: View Participation Analytics
:Administrator
:AnalyticsDashboard
navigateToParticipationAnalytics()
«display» Analytics Dashboard
(loaded from database)
selectMetric(e.g., "Registrations per Sport")
«display» Charts / Graphs for selected metric
applyDateRangeFilter(startDate, endDate)
alt
date range (start > end)
«display» Error: Invalid date range.
Ensure start date is before end date.
date range
«display» Refreshed Dashboard
for selected date range

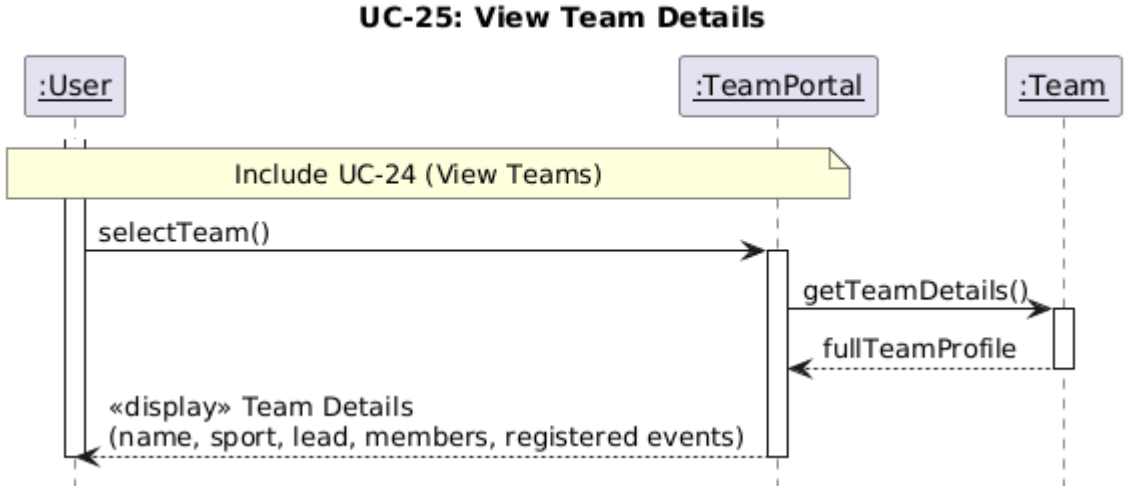
2.23 UC-23


UC-23: Manage Event
:Administrator
:EventManagement
:Event
navigateToEventManagement()
«display» List of Current and Upcoming Events
selectEvent()
getEventDetails()
eventDetails
«display» Event Details
(name, sport, date, registration status)
modifyEventParameters(params)
alt
date is in the past
«display» Error: Event date cannot be in the past.
date input
manageParticipants(addOrRemove)
alt
team name in same sport
«display» Error: Duplicate team name.
Cannot add team.
participant update
saveChanges()
«display» Success: Event updated.

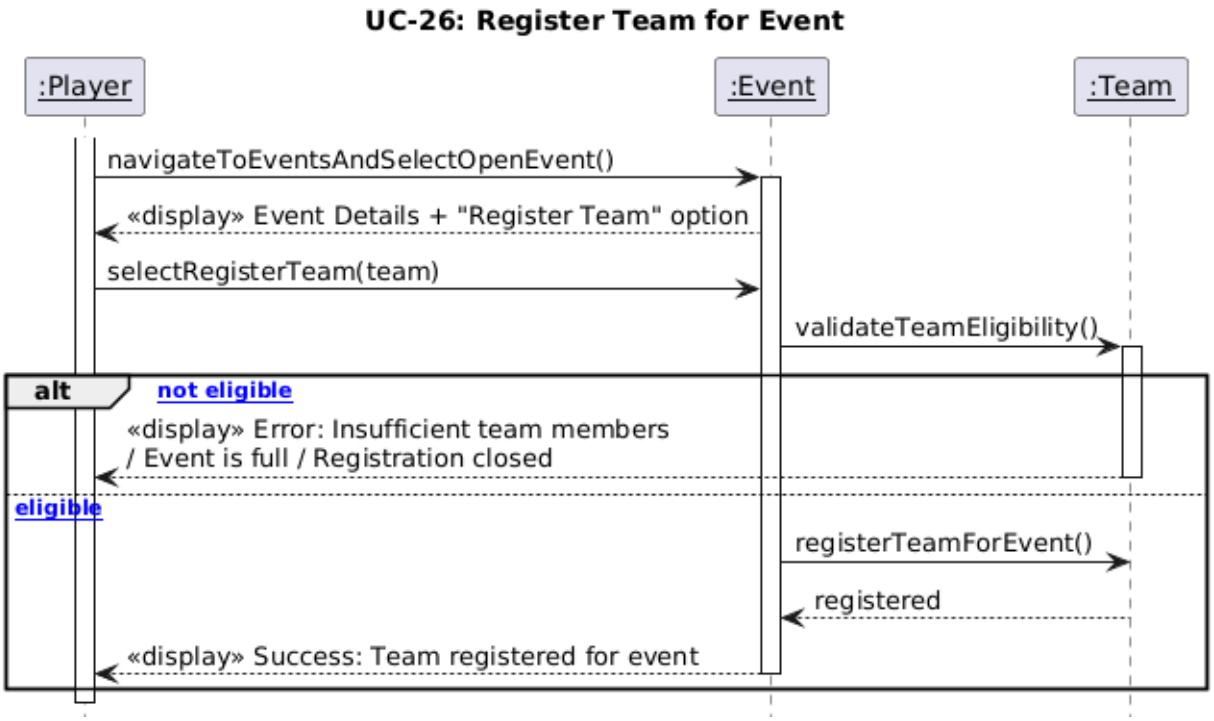
2.24 UC-24



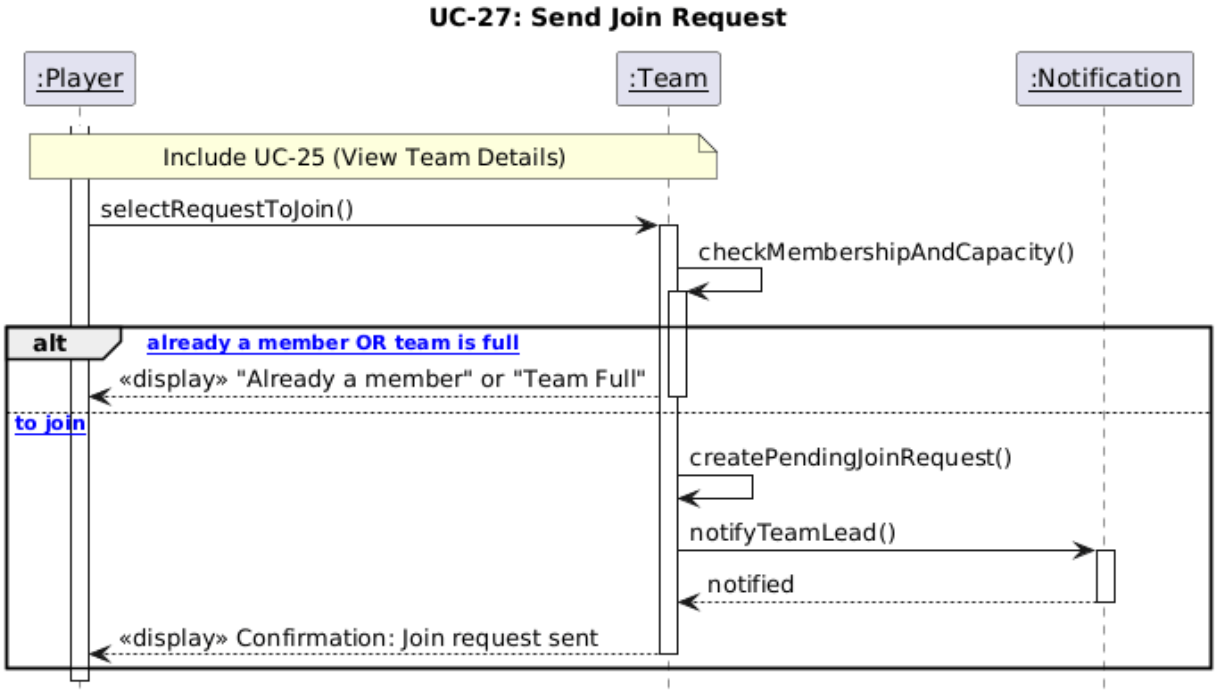
2.25 UC-25



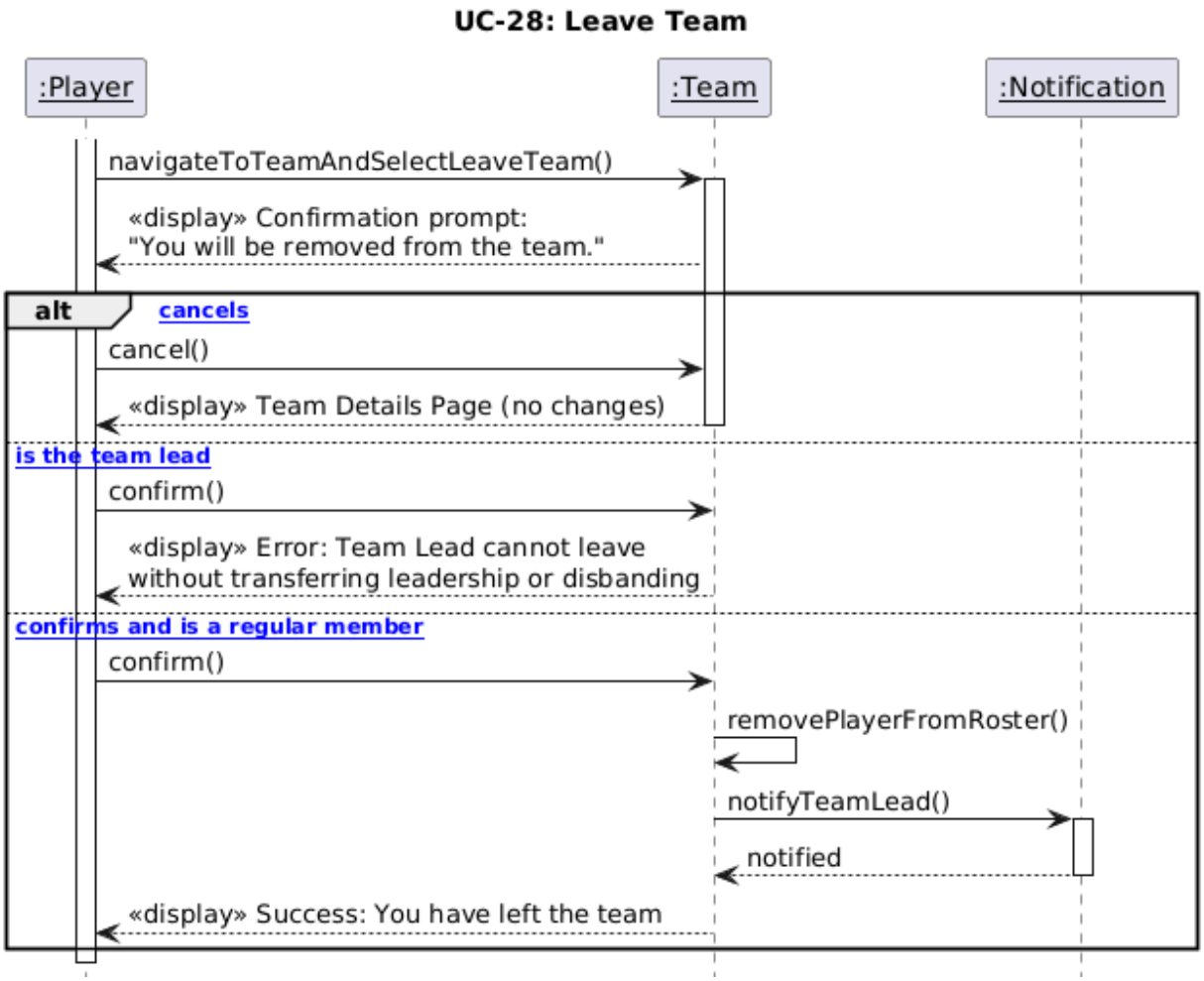
2.26 UC-26


UC-26: Register Team for Event
:Player
:Event
:Team
navigateToEventsAndSelectOpenEvent()
«display» Event Details + "Register Team" option
selectRegisterTeam(team)
validateTeamEligibility()
alt
not eligible
«display» Error: Insufficient team members
/ Event is full / Registration closed
eligible
registerTeamForEvent()
registered
«display» Success: Team registered for event

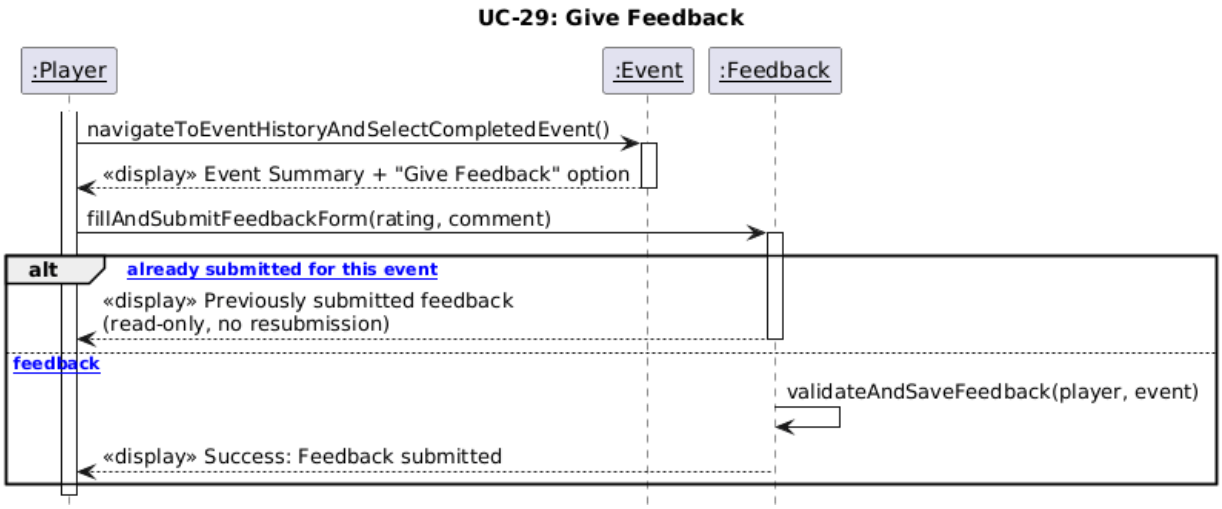
2.27 UC-27


UC-27: Send Join Request
:Player
:Team
:Notification
Include UC-25 (View Team Details)
selectRequestToJoin()
checkMembershipAndCapacity()
alt
already a member OR team is full
«display» "Already a member" or "Team Full"
to join
createPendingJoinRequest()
notifyTeamLead()
notified
«display» Confirmation: Join request sent

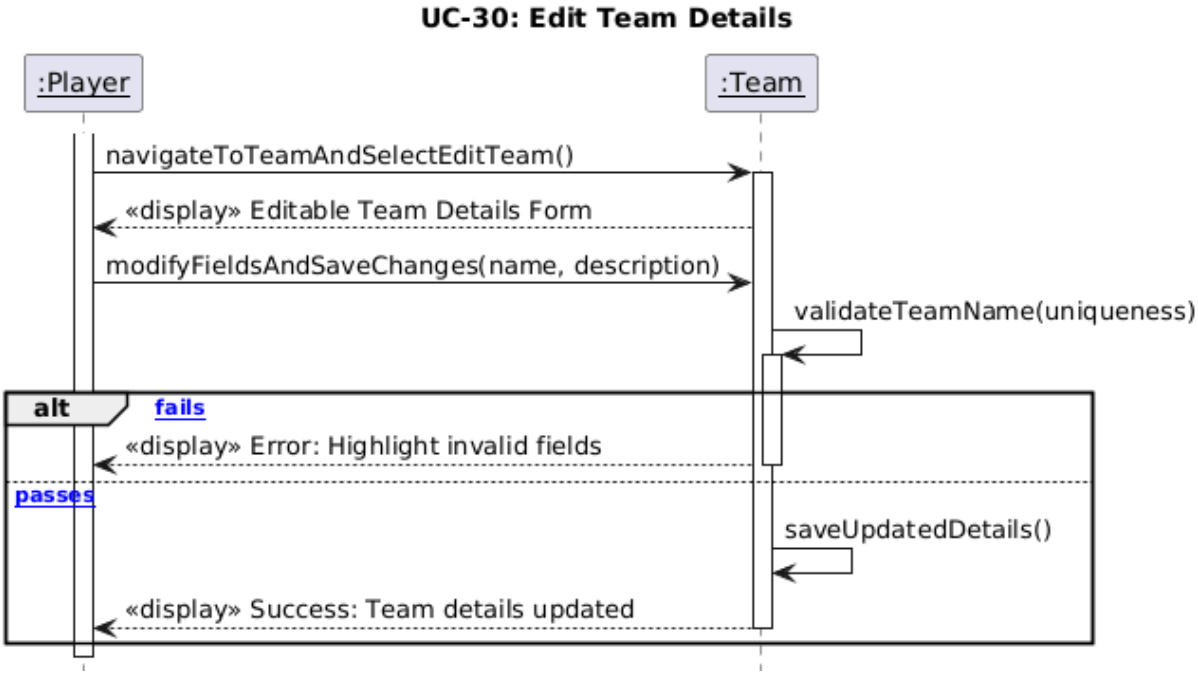
2.28 UC-28


UC-28: Leave Team
:Player
:Team
:Notification
navigateToTeamAndSelectLeaveTeam()
«display» Confirmation prompt:
"You will be removed from the team."
alt
cancels
cancel()
«display» Team Details Page (no changes)
is the team lead
confirm()
«display» Error: Team Lead cannot leave
without transferring leadership or disbanding
confirms and is a regular member
confirm()
removePlayerFromRoster()
notifyTeamLead()
notified
«display» Success: You have left the team

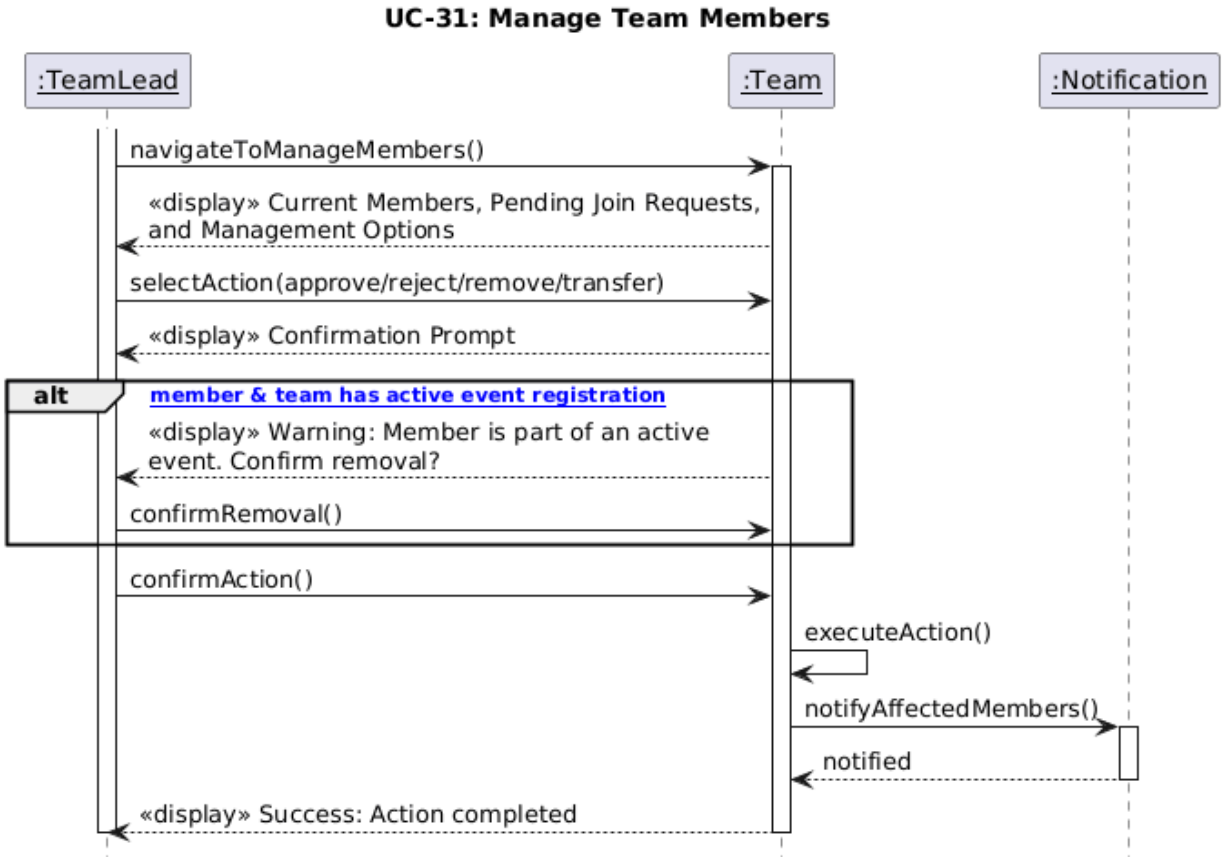
2.29 UC-29



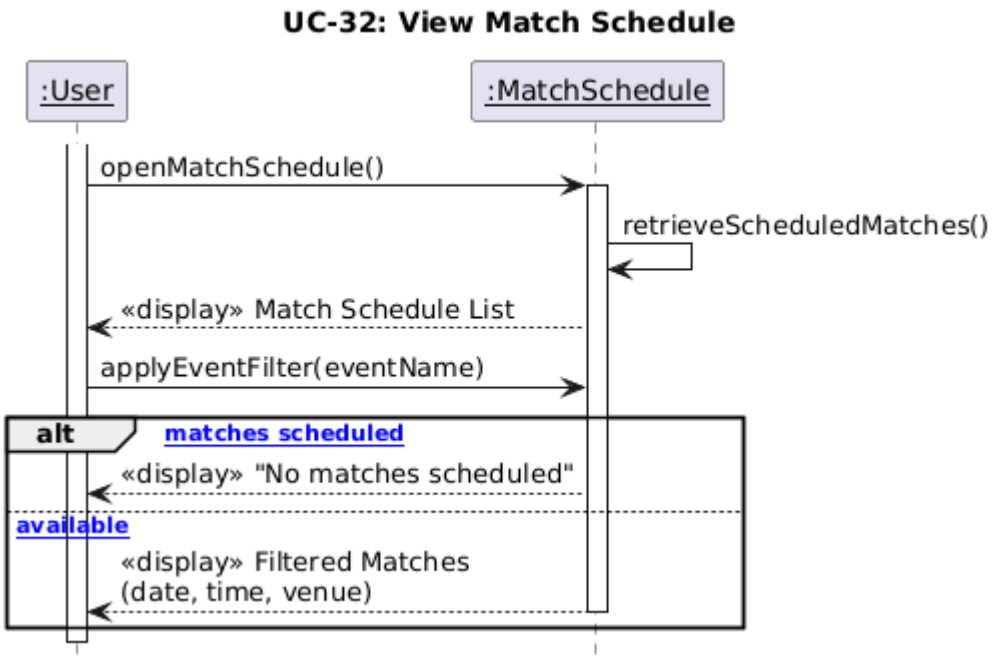
2.30 UC-30



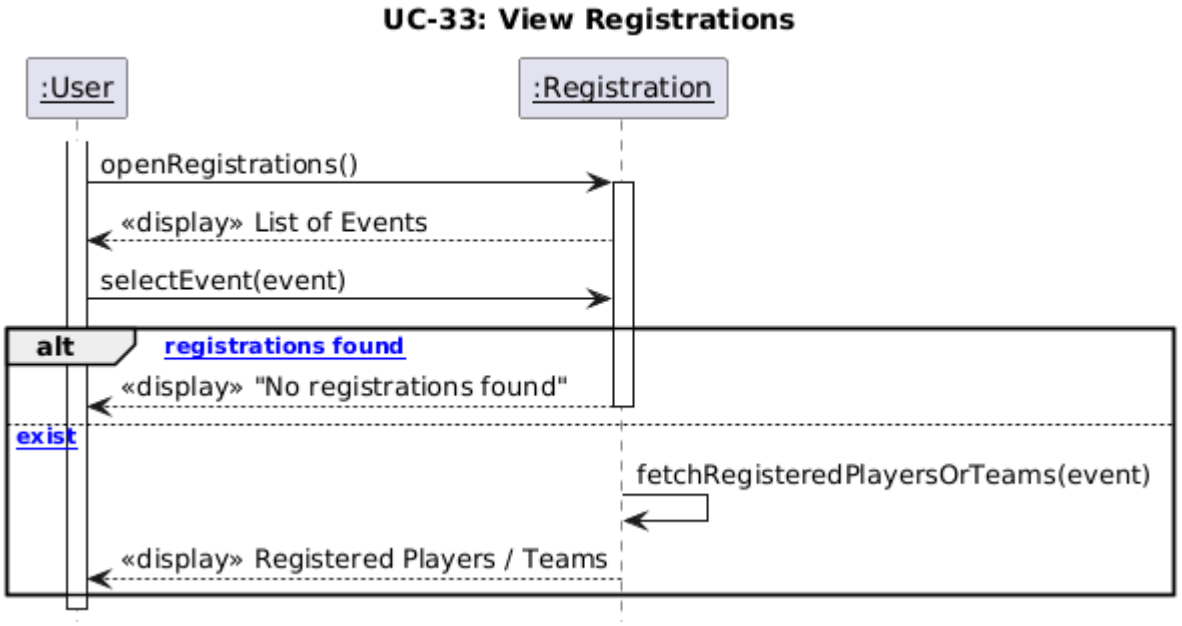
2.31 UC-31


UC-31: Manage Team Members
:TeamLead
:Team
:Notification
navigateToManageMembers()
«display» Current Members, Pending Join Requests, and Management Options
selectAction(approve/reject/remove/transfer)
«display» Confirmation Prompt
alt
member & team has active event registration
«display» Warning: Member is part of an active event. Confirm removal?
confirmRemoval()
confirmAction()
executeAction()
notifyAffectedMembers()
notified
«display» Success: Action completed

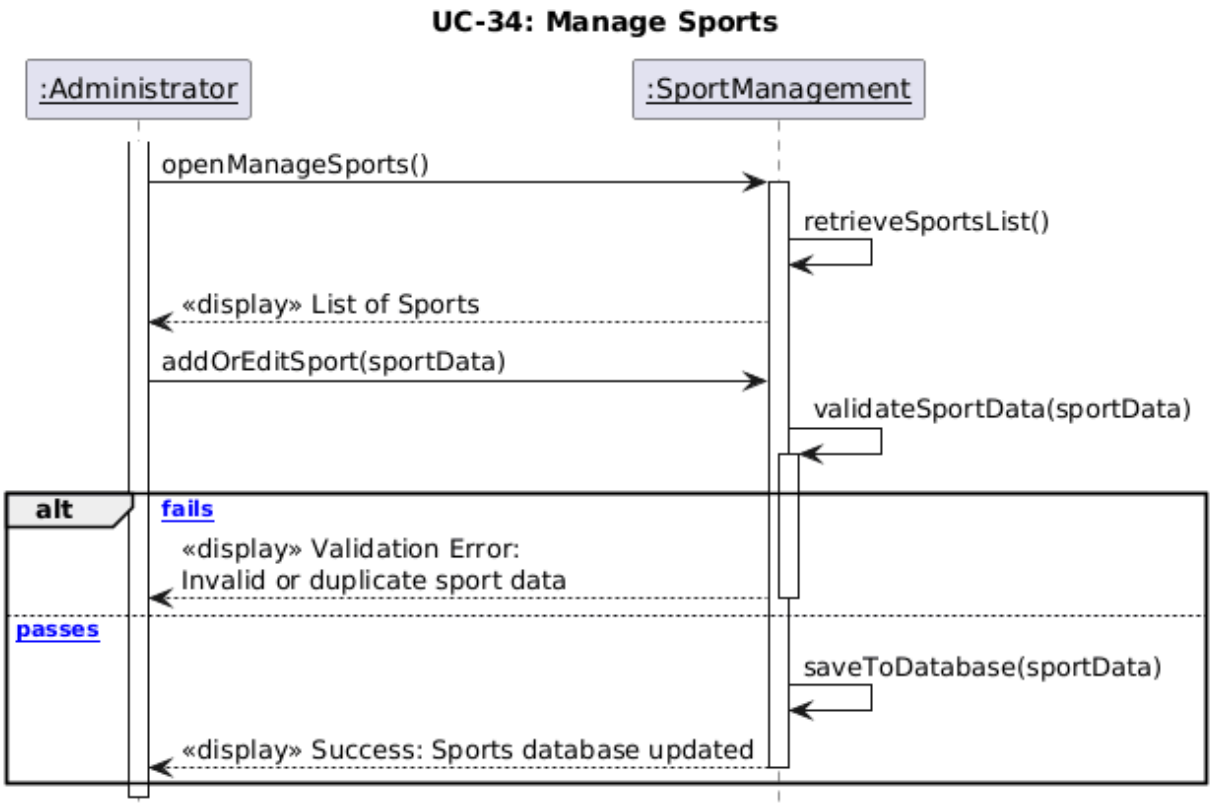
2.32 UC-32



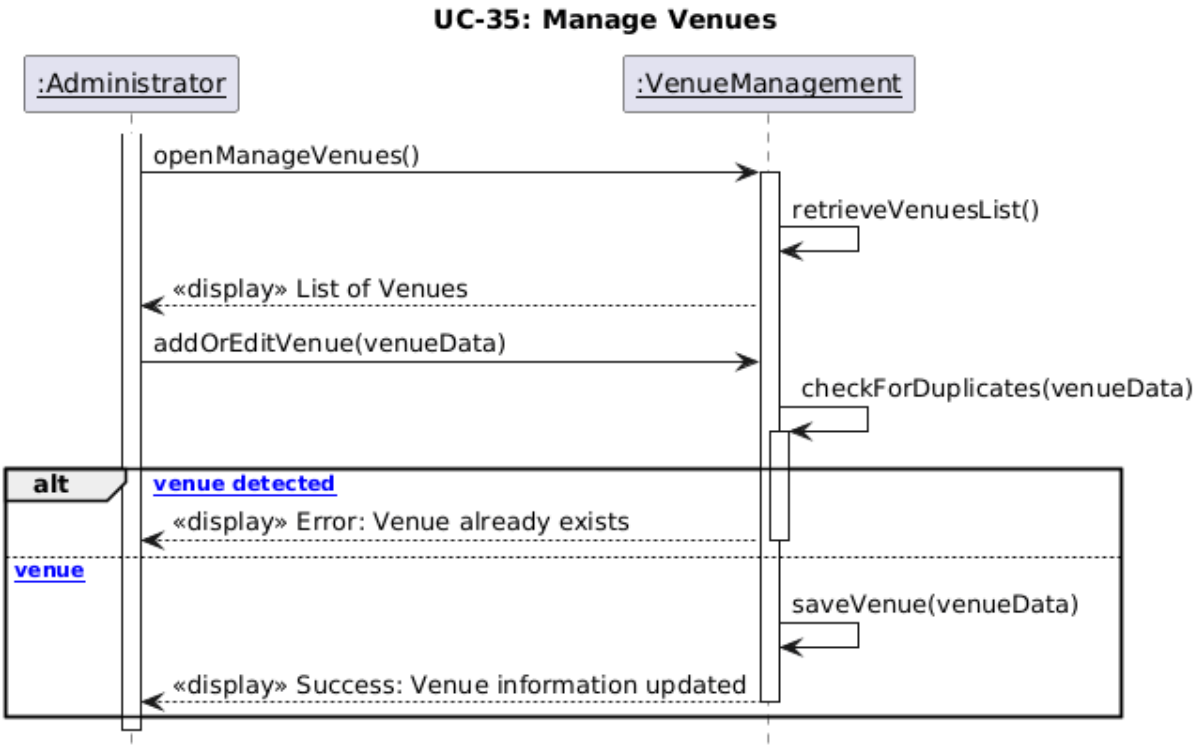
2.33 UC-33



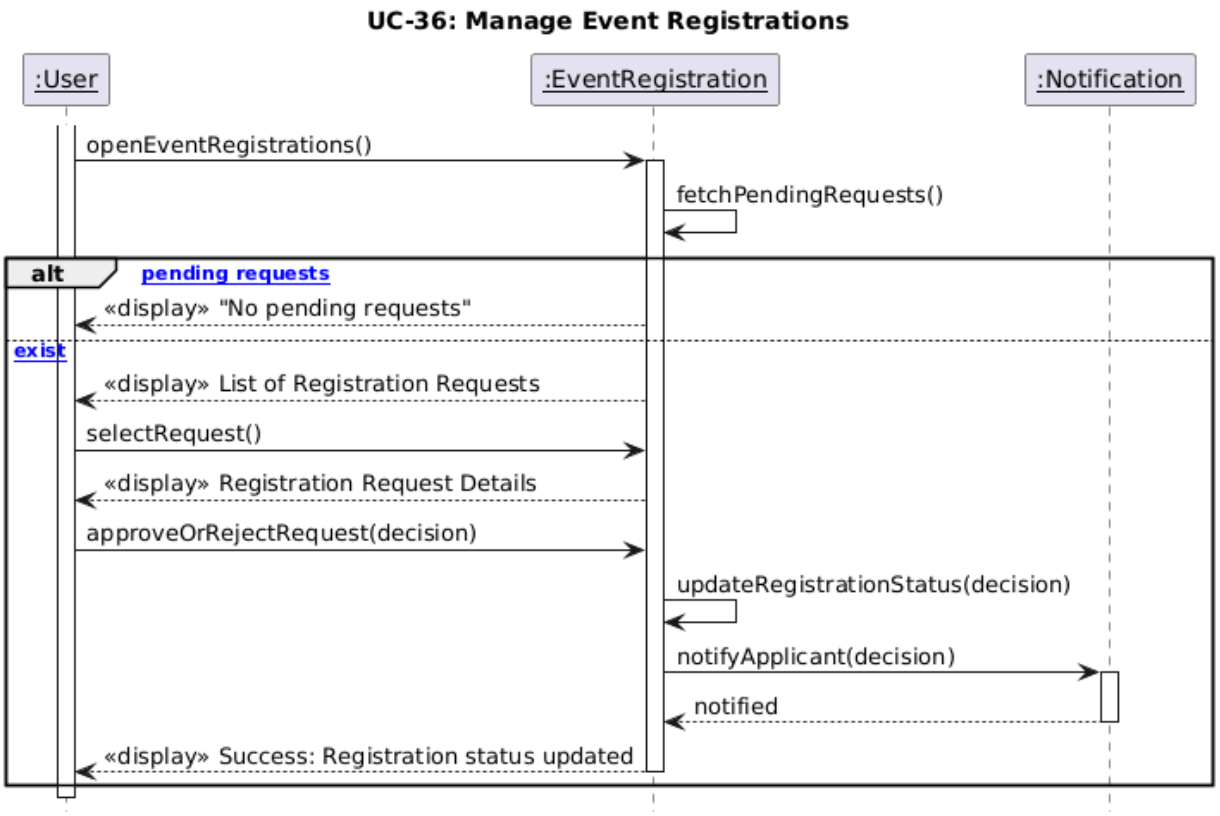
2.34 UC-34


UC-34: Manage Sports
:Administrator
:SportManagement
openManageSports()
retrieveSportsList()
«display» List of Sports
addOrEditSport(sportData)
validateSportData(sportData)
alt
fails
«display» Validation Error:
Invalid or duplicate sport data
passes
saveToDatabase(sportData)
«display» Success: Sports database updated

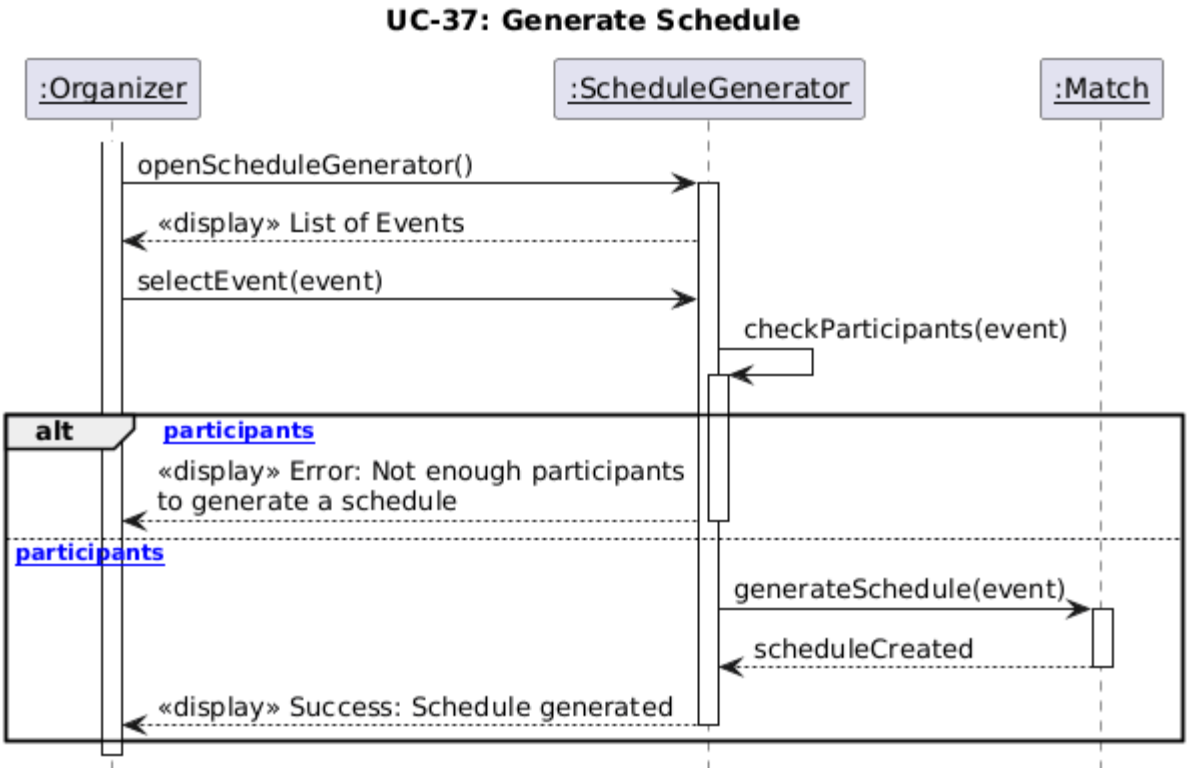
2.35 UC-35


UC-35: Manage Venues
:Administrator
:VenueManagement
openManageVenues()
retrieveVenuesList()
«display» List of Venues
addOrEditVenue(venueData)
checkForDuplicates(venueData)
alt
venue detected
«display» Error: Venue already exists
venue
saveVenue(venueData)
«display» Success: Venue information updated

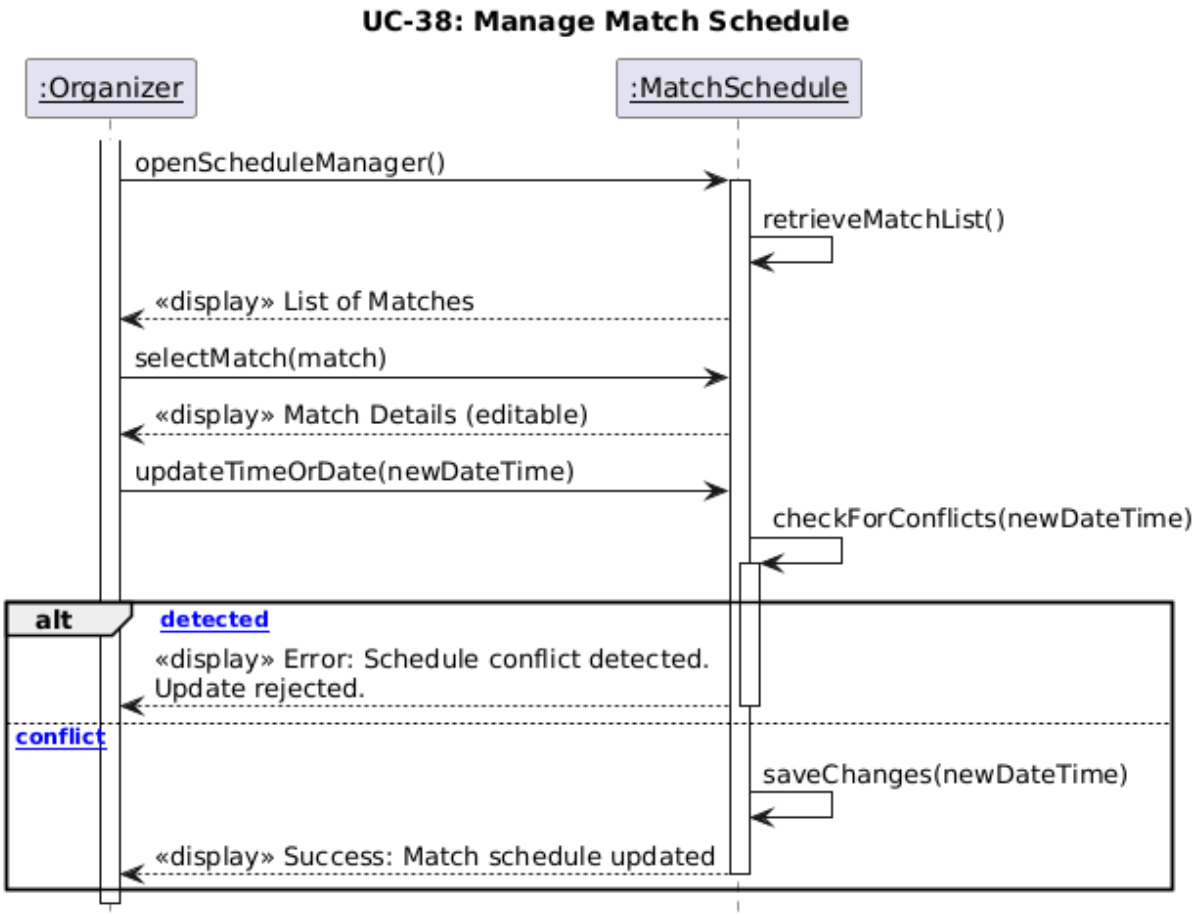
2.36 UC-36


UC-36: Manage Event Registrations
:User
:EventRegistration
:Notification
openEventRegistrations()
fetchPendingRequests()
alt
pending requests
«display» "No pending requests"
exist
«display» List of Registration Requests
selectRequest()
«display» Registration Request Details
approveOrRejectRequest(decision)
updateRegistrationStatus(decision)
notifyApplicant(decision)
notified
«display» Success: Registration status updated

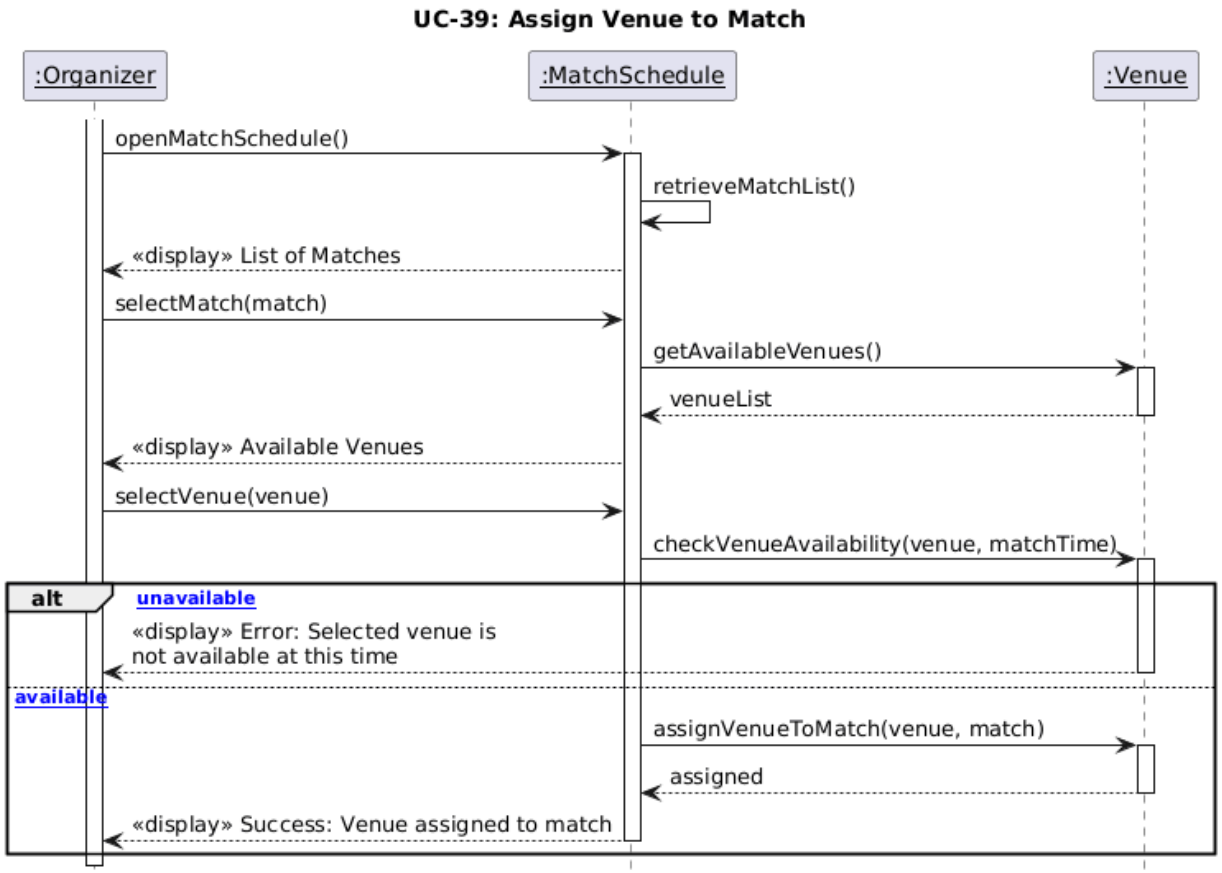
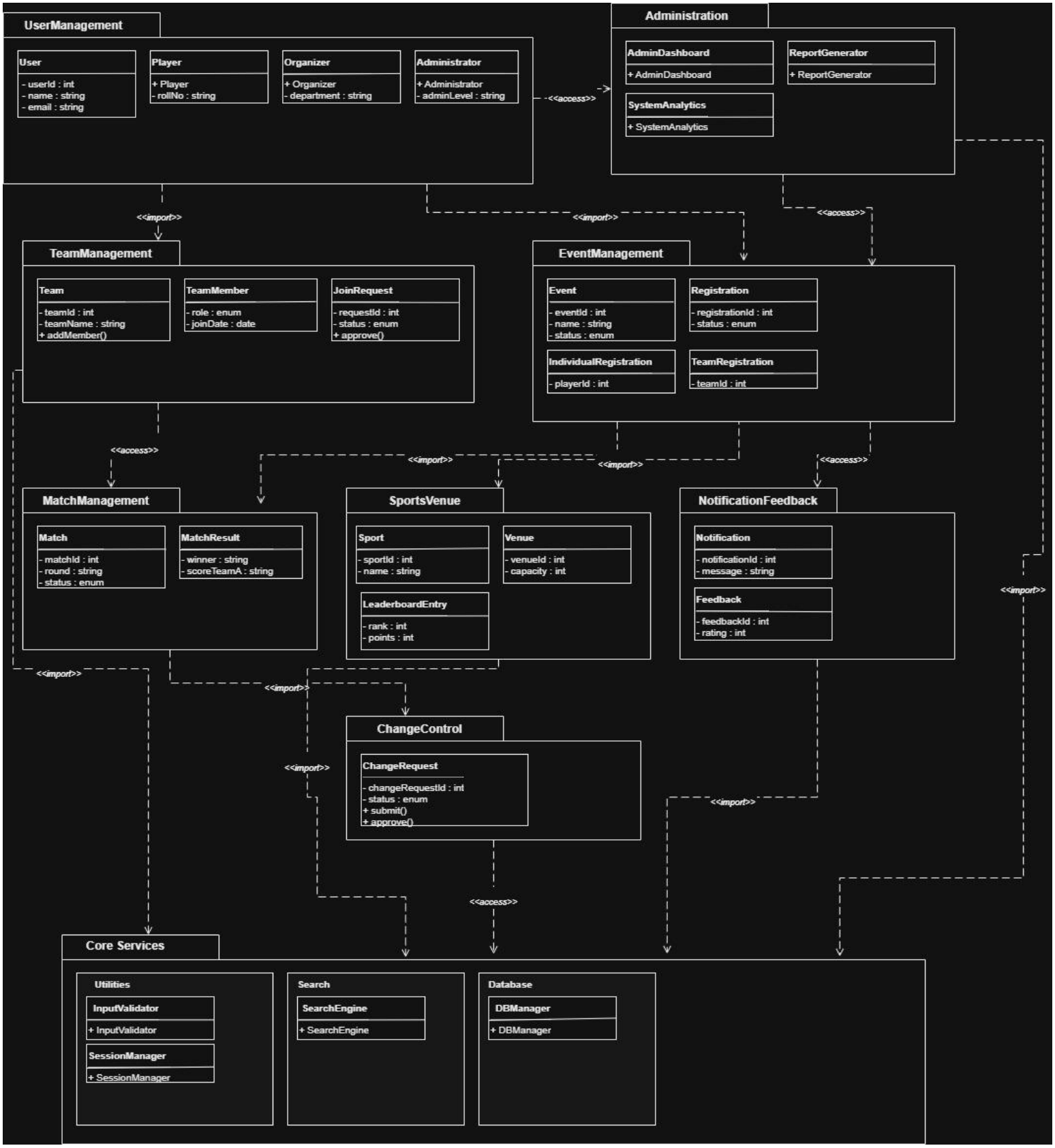
2.37 UC-37


UC-37: Generate Schedule
:Organizer
:ScheduleGenerator
:Match
openScheduleGenerator()
«display» List of Events
selectEvent(event)
checkParticipants(event)
alt
participants
«display» Error: Not enough participants to generate a schedule
participants
generateSchedule(event)
scheduleCreated
«display» Success: Schedule generated

2.38 UC-38


UC-38: Manage Match Schedule
:Organizer
:MatchSchedule
openScheduleManager()
retrieveMatchList()
«display» List of Matches
selectMatch(match)
«display» Match Details (editable)
updateTimeOrDate(newDateTime)
checkForConflicts(newDateTime)
alt
detected
«display» Error: Schedule conflict detected.
Update rejected.
conflict
saveChanges(newDateTime)
«display» Success: Match schedule updated

2.39 UC-39


UC-39: Assign Venue to Match
:Organizer
:MatchSchedule
:Venue
openMatchSchedule()
retrieveMatchList()
«display» List of Matches
selectMatch(match)
getAvailableVenues()
venueList
«display» Available Venues
selectVenue(venue)
checkVenueAvailability(venue, matchTime)
alt
unavailable
«display» Error: Selected venue is not available at this time
available
assignVenueToMatch(venue, match)
assigned
«display» Success: Venue assigned to match

3. Package Diagram: